

UMAR ASGHAR KHAN

MATERIALS & METALLURGICAL ENGINEER | OIL & GAS CORROSION & INSPECTION

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[LinkedIn](#):

CORE COMPETENCIES

Metallurgical Failure Analysis, Corrosion Engineering (CUI), Oil & Gas Inspection & Integrity, NDT (UT, PAUT, TOFD, MPI), Cathodic Protection Monitoring.

Professional Experience

MOL Pakistan Oil & gas company B.V

Inspection and Condition Monitoring Intern

2025

24 June 2025 – 5 August

- Assisted corrosion under insulation(CUI) inspection of mild steel process equipment.
- Conducted A-Scan Ultrasonic testing for thickness measured and integrity assessment.
- Applied PAUT and TOFD for weld inspection and defect characterization.
- Executed Magnetic Particle Inspection(MPI) and Holiday testing.
- Worked with Cathodic Protection (CP) system for buried pipelines.
- Used Thermography and Vibration Analysis for Condition monitoring.
- Assisted inspection engineers in API and ASME compliance activities.

Oil & Gas Development Company Limited (OGDCL)

Corrosion Intern

2024

20 July 2024 – 6 September

- Assisted corrosion inspection of pipelines, tanks, and static equipment.
- Conducted UT thickness measurements for corrosion rate assessment.
- Evaluated tank bottom -to- tank shell joints for corrosion damage.
- Supported corrosion engineers in integrity and inspection reporting.

Education

PAKISTAN INSTITUTE OF ENGINEERING & APPLIED SCIENCES

2022 - 2026

- BSc Materials & Metallurgical Engineering
- Key Courses: Corrosion Engineering, Physical Metallurgy, Surface Coatings, Welding Metallurgy, Materials Thermodynamics
- CGPA: 3.0 / 4.0

Projects

1. Final Year Project

Green Synthesized Banana Peel Carbon–CoO Composite Electrode for Super capacitors

Developed biomass-derived carbon electrodes using >2 kg banana peel precursor, incorporating 5–15 wt% CoO, achieving a maximum specific capacitance of ~300 F/g at 0.5 A/g.

Skills

Software: SolidWorks, MS Office, MS Excel, Procast, MS Word.

Programming: Python, C.

Electrochemical Testing: Cyclic voltammetry, Galvanic Static Charge and Discharge, Electrochemical impedance spectroscopy

Communication: Interpersonal skills, team management, work management, managerial skills.

Mechanical testing (tensile, hardness, impact, fatigue basics)

Heat treatment of steels and alloys

Failure analysis & root cause analysis

Corrosion testing and prevention

Non-Destructive Testing (UT, PAUT, TOFD, MPI, RT basics)

Welding metallurgy & weld inspection

Powder metallurgy & composite materials

Co-Curricular activities

- Volleyball Winner – Sports Week PIEAS 2024.
- Badminton Winner – PIEAS.
- HEC Volleyball team Captain – PIEAS.
- Volunteer Ramadan ration distribution (Muaawin Team).

Hobbies

Running, Hiking, Horse Riding.